

EXAMINATION INFORMATION PAGE

Written assignments, **Home exam**

Subject code: IIA4117	Subject name: Model Predictive Control	
Responsible subject teacher: Roshan Sharma	Campus: Porsgrunn	Faculty: TNM
Assignment given in WISEflow (date and time): 07.12.2021 Start time: 09:00		Submission time in WISEflow (date and time): 07.12.2021 End time: 12:30
No. of assignments: 3	No. of attachments: None	No. of pages incl. front page and attachments: 4

Aids and collaboration:

All aids are allowed.

	Yes	No
Is it an individual exam?	☒	☐
Is it allowed to collaborate with other persons?	☐	☒

Description of individual examination and illegal cooperation will be found at my.usn.no

Criteria for the answers:

Font type: Times New Roman Hand written solutions allowed	Font size: 12 Hand written solutions allowed	Line spacing: Single Hand written solutions allowed
No. of words (min/max): No limit	Maximum no. of pages excl. front page and attachments: No limit	

Other important information:

- i) Answer sheets written by hand are acceptable. The students should upload scanned copies of the answer sheets in pdf format, or take good quality pictures of the answer sheet (for example using a camera on your phone) and upload them. All the uploaded documents **MUST** be clearly legible i.e. the examiner should be able to read them properly. It is the responsibility of the students to upload answer sheets that are clearly readable.
- ii) It is strictly forbidden to communicate and collaborate, in any form, with other persons during the exam period.
- iii) This final written exam has a weight of 60% on the final grade. The remaining 40% weight on the final grade comes from the project work.
- iv) The exam ending time also includes the time that the students have to use for scanning (or taking pictures) and uploading the answer sheets in Wiseflow. After kl. 12:30, it will not be possible to upload answer sheets in Wiseflow. Please allocate enough time for submission part of your answer sheets.
- v) In case you think that a problem has a mistake or it is incomplete, explain the information you think it needs to be corrected or completed, assume values for this information if needed, and continue to solve the problem.

This exam counts 60% for the final grade.

Problem 1

Electrical submersible pump (ESP) lifting method is a way to extract heavy oil from an oil reservoir. For this a special type of pump called ESP are placed at the bottom of the well as shown in Figure 1. The pump is powerful enough to push the heavy oil from the reservoir all the way up to the top of the well. The top of the well is also known as well head.

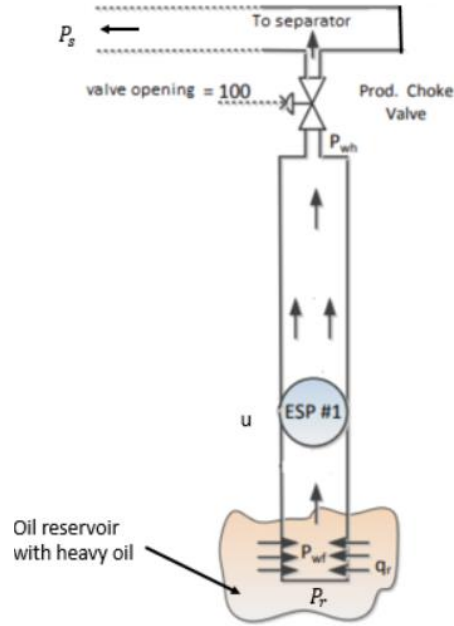


Figure 1: A schematic of an ESP lifted oil well

At the well head, there is a production choke valve which is usually kept fully open for a well that is in operation. How much oil is actually produced from the reservoir is dependent on various parameters of the well. One of the most important parameter is called the Gas-to-oil ratio (GOR). This parameter defines how much gas (compared to oil) is coming out of the well. For example, if $GOR = 0$, then it means only pure crude oil is flowing out of the well (which is what is desired). Higher values of GOR usually means that a lot of gas is also being produced together with the oil from a given well.

Thus to properly estimate the actual amount of oil flowing out from a well, the value of the parameter GOR should be properly known. And here lies the difficulty: In reality, GOR is a highly uncertain parameter. Uncertain in the sense that its actual value is quite difficult to measure and thus is usually not known exactly. Multiphase flow sensors help very little in this case, because such multiphase flow sensors perform poorly in a gassy oil field.

Now let us assume that you are hired as an automation engineer in an oil company working with ESP lifting method. You are given the responsibility to design a nonlinear MPC for optimizing (maximizing) the amount of oil produced from an oil field. For simplicity let us assume that there is only one oil well present in the field. Let us also assume that your manager provided you a discrete time dynamic model of the ESP lifted oil well as,

$$x_{k+1} = f(x_k, u_k, P_r, P_{wh}, P_s, GOR) \quad (1)$$

Here, x_k is the flow rate of oil from the well, u_k is the speed of the pump, P_r is the reservoir pressure, P_{wh} is the well head pressure, P_s is the separator pressure and GOR is the gas to oil ratio parameter. f is some nonlinear function describing the dynamics of x_k . Assume that all the pressure values are measurable and

known exactly at any given point of time. There are also some constraints that has to be fulfilled during the operation of the well.

$$x_{min} \leq x_k \leq x_{max} \quad (2)$$

$$u_{min} \leq u_k \leq u_{max} \quad (3)$$

The first constraint says that the amount of oil produced from the well should be at least x_{min} , and it should not exceed x_{max} . The second constraint says that the speed of the pump cannot be less than u_{min} , and it should not exceed u_{max} . The constraint on x_{max} is important since it is related to the maximum volumetric flow rate that the separator can handle. In other words, we cannot produce more than x_{max} amount of oil from the well, otherwise the separator capacity will be reached and overflow might occur.

Since the parameter GOR is uncertain, let us assume that your manager did not give you a fixed (constant) value for GOR . Instead the manager said to you that GOR can take any values between 0.1 and 0.5 i.e. $GOR \in [0.1 \ 0.5]$.

As a first try, let us assume that you selected a nominal value for the parameter i.e. $GOR = 0.3$, and then designed a nominal MPC with the objective of maximizing the oil production + constraint satisfaction. When you tested your MPC considering that the value of $GOR = 0.3$ for the real plant (target system, see Figure 2a), it worked satisfactorily. However, your manager then reminded you that for the real plant, GOR could take any values between 0.1 and 0.5. Thus you decided to test your nominal MPC against the real plant (target system, see Figure 2b) with $GOR = 0.4$. To your surprise, you found out that your nominal MPC did not work satisfactorily and the constraint on x_{max} was violated.

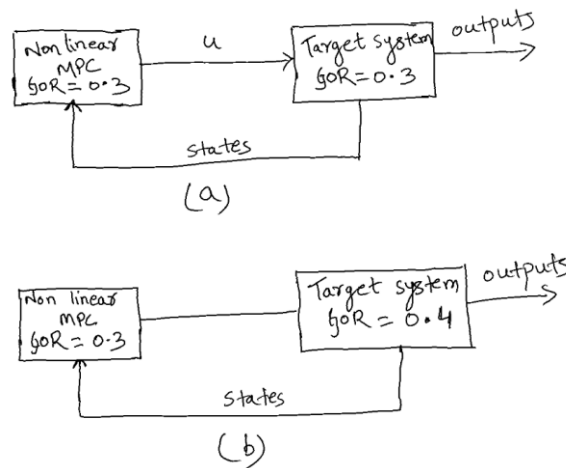


Figure 2: Nominal MPC for target system with different values of the same parameter.

To handle this, you decided to classify the parameter GOR into 5 values such that $GOR = [0.1, 0.2, 0.3, 0.4, 0.5]$, and you decided to use multi objective optimization methods to design a more robust MPC. Let us assume that the model of ESP lifted oil well was re-written to take into account all the five values for GOR as,

$$x_{k+1}^i = f^i(x_k^i, u_k, P_r, P_{wh}^i, P_s, GOR^i) \quad (4)$$

Here, the superscript i for different variables correspond to the i^{th} value of the parameter GOR .

Tasks

- (i) [20%] Explain and show in detail, how you would formulate nonlinear MPC so that you take into account all the 5 possible values of the parameter GOR . Show your objective function, constraints etc.
- (ii) [5%] Assuming that the constraint on u_{min} is a soft constraint. How would you re-formulate the nonlinear optimal control problem to handle infeasibility that might arise due to constraint on

u_{min} . Please also note that the minimum speed of the pump cannot be less than zero (since zero pump speed means that the pump is already completely stopped).

Problem 2

With the Kronecker product formulation of an LQ optimal control problem, the size of the control problem can easily become very large. The size of the control problem can be reduced by elimination of the equality constraints.

Let us consider an LQ optimal control problem expressed in the standard QP form as,

$$\left. \begin{array}{l} \min_z \frac{1}{2} z^T H z + c^T z \\ \text{subject to} \\ A_e z = b_e \\ A_i z \leq b_i \\ z_L \leq z \leq z_H \end{array} \right\} \quad (5)$$

The problem given by equation (5) can be converted into a QP problem of the reduced form as,

$$\left. \begin{array}{l} \min_{z_2} \frac{1}{2} z_2^T \tilde{H} z_2 + \tilde{c}^T z_2 \\ \tilde{A}_i z_2 \leq \tilde{b}_i \end{array} \right\} \quad (6)$$

Tasks

- (i) [15%] Explain and show in detail how QR factorization can be used to eliminate the equality constraints in order to reduce the size of the optimal control problem. Include all necessary calculations and equations. In addition, please show the detailed intermediate steps of how expressions for $\tilde{H}$ and $\tilde{c}$ are calculated.

Note: For Problem 2, task (i), you cannot just say “after simplifying and re-arranging equation...” we get the expression for $\tilde{H}$ and $\tilde{c}$ and then only write down/copy the final expressions for $\tilde{H}$ and $\tilde{c}$. You should also do and show your own calculations. Students who have only written the final expression for $\tilde{H}$ and $\tilde{c}$ without showing the intermediate calculations/steps will not receive full score.

- (ii) [5%] Will you use optimization problem given by Equation (5) or that given by Equation (6) to develop an LQ optimal controller for embedded systems? Give proper explanation.

Problem 3

Tasks

- (i) [5%] Describe the concept about warm start when solving an optimization problem recursively/repeatedly. What is its advantage?
- (ii) [5%] For a real industrial process, a state feedback MPC will seldom work. Do you agree with this statement? If yes why? If no why?
- (iii) [5%] In practice, during the implementation of MPC for an open loop stable system, the terminal constraints and the terminal cost are usually discarded. Do you agree with this statement? Please justify your answer.